

Containerize Springboot+MySQL

01 August 2026 20:59

Step1: Create a springboot project

```
SpringBootMySQL/
├── src/
│   ├── main/
│   │   ├── java/
│   │   │   ├── com/example/demo/
│   │   │   │   ├── DemoApplication.java
│   │   │   │   ├── controller/
│   │   │   │   │   ├── StudentController.java
│   │   │   │   │   ├── entity/
│   │   │   │   │   │   ├── Student.java
│   │   │   │   │   ├── repository/
│   │   │   │   │   │   ├── StudentRepository.java
│   │   │   │   │   ├── service/
│   │   │   │   │   │   ├── StudentService.java
│   │   └── resources/
│   │       ├── application.properties
│   ├── Dockerfile
│   ├── docker-compose.yml
│   └── pom.xml
```

For creating Dockerfile and docker-compose.yml we need to right click on project and click the create file and create them.

DockerFile:

```
FROM eclipse-temurin:17-jre

WORKDIR /app

COPY target/*.jar app.jar

EXPOSE 8181

ENTRYPOINT ["java", "-jar", "app.jar"]
```

docker-compose.yml:

```
version: "3.8"

services:

  mysql:

    image: mysql:8.4

    container_name: mysql-db

    restart: always

    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: studentdata

    ports:
      - "3306:3306"

    volumes:
      - mysql-data:/var/lib/mysql

  springboot:

    build: .

    container_name: springboot-app

    depends_on:
      - mysql

    ports:
      - "8181:8181"

    environment:
      SPRING_DATASOURCE_URL: jdbc:mysql://mysql:3306/studentdata
      SPRING_DATASOURCE_USERNAME: root
      SPRING_DATASOURCE_PASSWORD: root

    volumes:
      mysql-data:
```

Student.java:

```
package com.ContainerizeSpringMySQL.ContainerizeSPMY.entity;

import jakarta.persistence.*;

@Entity
@Table(name="students")
public class Student {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)

    private Long id;

    private String name;

    private String course;

    private int age;

    public Student() {
    }

    public Student(String name, String course, int age) {
        this.name = name;
        this.course = course;
        this.age = age;
    }

    public Long getId() {
        return id;
    }
}
```

```

    public void setId(Long id){
        this.id=id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name){
        this.name=name;
    }

    public String getCourse() {
        return course;
    }

    public void setCourse(String course){
        this.course=course;
    }

    public int getAge() {
        return age;
    }

    public void setAge(int age){
        this.age=age;
    }
}

```

StudentRepository.java:

```

package com.ContainarizeSpringMySQL.ContanarizeSPMY.repository;

import org.springframework.data.jpa.repository.JpaRepository;

import com.ContainarizeSpringMySQL.ContanarizeSPMY.entity.Student;

public interface StudentRepository extends JpaRepository<Student, Long> {

}

```

StudentService.java:

```

package com.ContainarizeSpringMySQL.ContanarizeSPMY.service;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import com.ContainarizeSpringMySQL.ContanarizeSPMY.entity.Student;
import com.ContainarizeSpringMySQL.ContanarizeSPMY.repository.StudentRepository;

@Service
public class StudentService {

    @Autowired
    StudentRepository repository;

    public Student saveStudent(Student student) {
        return repository.save(student);
    }

    public List<Student> getStudents() {
        return repository.findAll();
    }

    public Student getStudent(Long id) {
        return repository.findById(id).orElse(null);
    }

}

```

StudentController.java:

```

package com.ContainarizeSpringMySQL.ContanarizeSPMY.controller;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;

import com.ContainarizeSpringMySQL.ContanarizeSPMY.entity.Student;
import com.ContainarizeSpringMySQL.ContanarizeSPMY.service.StudentService;

@RestController
@RequestMapping("/students")
public class StudentController {

    @Autowired
    private StudentService service;

    @PostMapping("/add")
    public Student saveStudent(@RequestBody Student student) {
        return service.saveStudent(student);
    }

    @GetMapping("/all")
    public List<Student> getStudents() {
        return service.getStudents();
    }

    @GetMapping("/find/{id}")
    public Student getStudent(@PathVariable Long id) {
        return service.getStudent(id);
    }

}

```

Application.properties:

```

spring.application.name=ContanarizeSPMY

server.port=8181

spring.datasource.url=jdbc:mysql://localhost:3306/studentdata
spring.datasource.username=root
spring.datasource.password=root

spring.jpa.hibernate.ddl-auto=update

```

```
spring.jpa.show-sql=true
spring.jpa.database-platform=org.hibernate.dialect.MySQLDialect

Create a (.jar) -> right click on the project and run -> run build

Step2:Open the postman and check the /get and /post
```

Step3:cd C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY

Step4:Build the image docker build -t containarizespm .
And then click docker images
and then we include docker compose up --build to run multiple containers at once

Step5:Run Mysql Container
docker exec -it mysql-docker bash -> to run the command directly on the cmd bash
docker run -d --name mysql-container -e MYSQL_ROOT_PASSWORD=root -e MYSQL_DATABASE=studentdata -p 3306:3306 mysql:8.4

Step6:Run Springboot Container
docker run -d --name spring-container -p 8181:8181 containarizespm

Step7: Check the running Containers
docker ps

Step8:Check the logs
docker logs spring-container
docker logs mysql-container

Step9: Check the apis /get and /post if any other

Step10:
docker stop spring-container
docker stop mysql-container

Step11:
docker rm -f spring-container
docker rm -f mysql-container

Step12: Image contains the .jar that we have created so we delete the image and .jar at once!!
docker rmi containarizespm

Step Necessary when the docker/eclipse uses the same port:

Check what is using port 3306

Run:

```
netstat -ano | findstr :3306
You'll get something like:
```

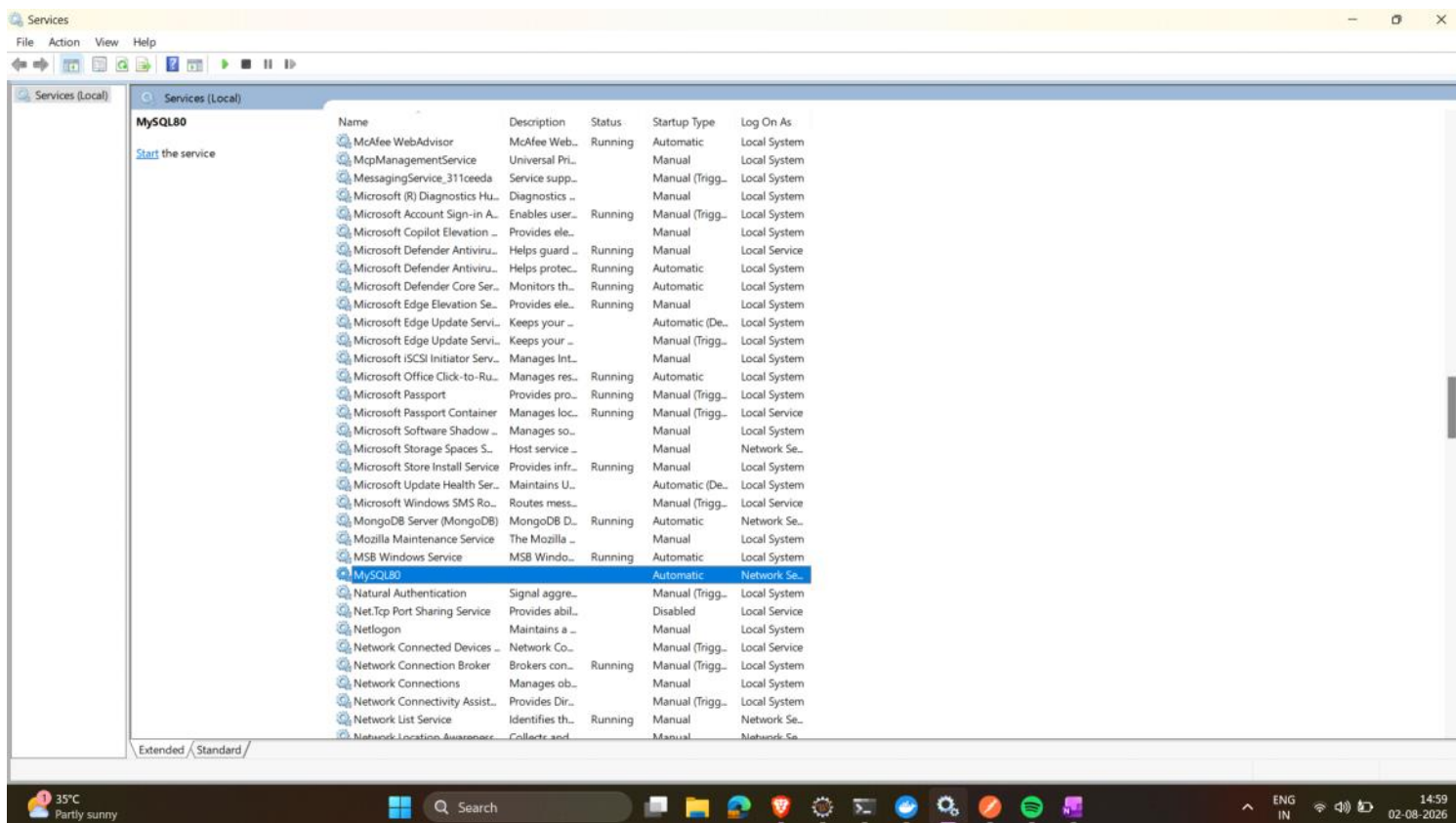
```
TCP 0.0.0.0:3306 0.0.0.0:0 LISTENING 12345
Then run:
```

```
tasklist /FI "PID eq 12345"
Replace 12345 with your PID.
```

Option 1 (Recommended): Stop Local MySQL

If it's your Windows MySQL service:
Open **Services** (Win + R -> services.msc)
Find:

MySQL80
Or



MySQL
Right-click -> Stop
Then run:

```
docker run -d --name mysql-container -e MYSQL_ROOT_PASSWORD=root123 -e MYSQL_DATABASE=studentdata -p 3306:3306 mysql
```

Errors that you face while creating the containers:

```
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker run -d --name spring-container -p 8181:8181
containarizespm
4260c242c8cdd37b9bb74e47b58af22b4d2427ab203147114ced9047513bd415
```

What's next:

Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:8181 -> 127.0.0.1:0: listen tcp 0.0.0.0:8181: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.

Run 'docker run --help' for more information

The error is still the same:

ports are not available:

0.0.0.0:8181

This is **not** a Docker image problem. It means **something on your Windows machine is already using port 8181**.

Step 1: Find the process using port 8181

Run this command:

```
netstat -ano | findstr :8181
```

Paste the output.

Step 2: Identify the process

Suppose the output is:

```
TCP 0.0.0.0:8181 0.0.0.0:0 LISTENING 12345
```

Then run:

```
tasklist /FI "PID eq 12345"
```

Replace 12345 with the PID from your output.

Step 3: Free the port

If it's your Spring Boot application running from Eclipse, stop it in Eclipse.

Or kill it:

```
taskkill /PID 12345 /F
```

Step 4: Run the container again

```
docker run -d --name spring-container -p 8181:8181 containarizespm
```

Perfect! We found the problem.

```
TCP 0.0.0.0:8181 LISTENING 5140
```

PID 5140 is already using port 8181. That's why Docker cannot use it.

Step 1: Find what PID 5140 is

Run:

```
tasklist /FI "PID eq 5140"
```

If it shows:

```
java.exe
```

then your **Spring Boot application is still running from Eclipse**.

Step 2: Stop it

Option A (Recommended)

In Eclipse:

- Go to the **Console** view.
- Click the **red square (Stop)** button.

OR

Option B

Kill the process:

```
taskkill /PID 5140 /F
```

Step 3: Verify the port is free

Run:

```
netstat -ano | findstr :8181
```

There should be **no output**.

Step 4: Remove the old container (if it exists)

```
docker rm -f spring-container
```

Step 5: Run the container again

```
docker run -d --name spring-container -p 8181:8181 containarizespm
```

Step 6: Check that it started

```
docker ps
```

You should see something like:

```
CONTAINER ID IMAGE STATUS
```

```
xxxxxx containarizespm Up 5 seconds
```

Then check the logs:

```
docker logs spring-container
```

You should see:

```
Started ContanarizeSpmApplication
```

```
Tomcat started on port 8181
```

If you don't want to stop Eclipse, use a different host port:

```
docker run -d --name spring-container -p 8085:8181 containarizespm
```

Then open:

<http://localhost:8181/students/all>

One more thing

Please run:

tasklist /FI "PID eq 5140"

PowerShell commands :

```
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker build -t containerizespm .
[+] Building 2.3s (9/9) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 164B 0.0s
=> [internal] load metadata for docker.io/library/eclipse-temurin:17-jre 1.9s
=> [auth] library/eclipse-temurin:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/3] FROM docker.io/library/eclipse-temurin:17-jre@sha256:92999aea37688157a53a40bfc187c30f317422e028045fd5f 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 91B 0.0s
=> CACHED [2/3] WORKDIR /app 0.0s
=> CACHED [3/3] COPY target/*.jar app.jar 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:9ae31a1f5f699203cf65209a7d976ddfd881453f524254f708c1126a84ae3c07 0.0s
=> => naming to docker.io/library/containerizespm 0.0s
```

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/xa239851115e0hrmtwwdcoa00

```
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker images
i Info -> U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
containerizespm:latest 9ae31a1f5f69 347MB 0B
mysql:8.4 bced325a4ab7 813MB 0B
nginx:latest 4e5db4761e0f 161MB 0B U
nginx:stable-alpine3.24-perl 64788b97f1a5 103MB 0B U
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker run -d --name mysql-container -e
MYSQL_ROOT_PASSWORD=Kingdeepak!@!17 -e MYSQL_DATABASE=studentdata -p 3306:3306 mysql:8.4
934d0070b5df0b48236c634f62aa00134b9b9871f1c949f428a64167d0b1afe8
```

What's next:
Debug this container error with Gordon -> docker ai "help me fix this container error"
docker: Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:3306 -> 127.0.0.1:0: listen tcp 0.0.0.0:3306: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.

```
Run 'docker run --help' for more information
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker run -d --name mysql-container -e
MYSQL_ROOT_PASSWORD=Kingdeepak!@!17 -e MYSQL_DATABASE=studentdata -p 3306:3306 mysql:8.4
```

What's next:
Debug this container error with Gordon -> docker ai "help me fix this container error"
docker: Error response from daemon: Conflict. The container name "/mysql-container" is already in use by container "934d0070b5df0b48236c634f62aa00134b9b9871f1c949f428a64167d0b1afe8". You have to remove (or rename) that container to be able to reuse that name.

```
Run 'docker run --help' for more information
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
934d0070b5df mysql:8.4 "docker-entrypoint.s..." About a minute ago Created mysql-container
e1439c60987c nginx "/docker-entrypoint..." 26 hours ago Exited (0) 26 hours ago gifted_shtern
e5d68b44184a nginx:stable-alpine3.24-perl "/docker-entrypoint..." 26 hours ago Exited (0) 26 hours ago
determined_buck
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker rm -f mysql-container
mysql-container
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker run -d --name mysql-container -e
MYSQL_ROOT_PASSWORD=Kingdeepak!@!17 -e MYSQL_DATABASE=studentdata -p 3306:3306 mysql:8.4
fb22b9506ee8320a92c0b952000d0fd2a9b6f2158f923b70050dbe4ad5667e15
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker run -d --name spring-container -p 8181:8181
containerizespm
4260c242c8cd37b9bb74e47b58af22b4d2427ab203147114ced9047513bd415
```

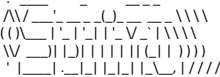
What's next:
Debug this container error with Gordon -> docker ai "help me fix this container error"
docker: Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:8181 -> 127.0.0.1:0: listen tcp 0.0.0.0:8181: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.

```
Run 'docker run --help' for more information
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
fb22b9506ee8 mysql:8.4 "docker-entrypoint.s..." 51 seconds ago Up 50 seconds 0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp
mysql-container
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker stop spring-container
spring-container
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker rm spring-container
spring-container
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker run -d --name spring-container -p 8181:8181
containerizespm
6aaf76fa193ca0fb422ec4c0a9f7d5ae64a4c2664c53a24f4a7f7f8d99bbcb64
```

What's next:
Debug this container error with Gordon -> docker ai "help me fix this container error"
docker: Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:8181 -> 127.0.0.1:0: listen tcp 0.0.0.0:8181: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.

```
Run 'docker run --help' for more information
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> netstat -ano | findstr :8181
TCP 0.0.0.0:8181 0.0.0.0:0 LISTENING 5140
TCP [::]:8181 [::]:0 LISTENING 5140
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> tasklist /FI "PID eq 5140"
```

```
Image Name PID Session Name Session# Mem Usage
=====
javaw.exe 5140 Console 35 2,90,588 K
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> taskkill /PID 5140 /F
SUCCESS: The process with PID 5140 has been terminated.
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> netstat -ano | findstr :8181
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker rm -f spring-container
spring-container
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker run -d --name spring-container -p 8181:8181
containerizespm
a0dd9def7e842a7415aea0ade0584b6fc35f863e2a1ccd17d2182822f84304fb
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
a0dd9def7e84 containerizespm "java -jar app.jar" 8 seconds ago Up 7 seconds 0.0.0.0:8181->8181/tcp, [::]:8181->8181/tcp
spring-container
fb22b9506ee8 mysql:8.4 "docker-entrypoint.s..." 2 minutes ago Up 2 minutes 0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp
mysql-container
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker logs spring-container
```



=====|_|=====|_|=|_|_|_|

:: Spring Boot :: (v4.1.0)

```
2026-08-02T08:26:07.134Z INFO 1 --- [ContanarizeSPMY] [ main] c.C.C.ContanarizeSpmApplication : Starting
ContanarizeSpmApplication v0.0.1-SNAPSHOT using Java 17.0.19 with PID 1 (/app/app.jar started by root in /app)
2026-08-02T08:26:07.139Z INFO 1 --- [ContanarizeSPMY] [ main] c.C.C.ContanarizeSpmApplication : No active profile set,
falling back to 1 default profile: "default"
2026-08-02T08:26:08.183Z INFO 1 --- [ContanarizeSPMY] [ main] s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring
Data JPA repositories in DEFAULT mode.
2026-08-02T08:26:08.251Z INFO 1 --- [ContanarizeSPMY] [ main] s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data
repository scanning in 55 ms. Found 1 JPA repository interface.
2026-08-02T08:26:08.999Z INFO 1 --- [ContanarizeSPMY] [ main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with
port 8181 (http)
2026-08-02T08:26:09.023Z INFO 1 --- [ContanarizeSPMY] [ main] o.apache.catalina.core.StandardService : Starting service
[Tomcat]
2026-08-02T08:26:09.024Z INFO 1 --- [ContanarizeSPMY] [ main] o.apache.catalina.core.StandardEngine : Starting Servlet
engine: [Apache Tomcat/11.0.22]
2026-08-02T08:26:09.093Z INFO 1 --- [ContanarizeSPMY] [ main] b.w.c.s.WebApplicationContextInitializer : Root
WebApplicationContext: initialization completed in 1859 ms
2026-08-02T08:26:09.525Z INFO 1 --- [ContanarizeSPMY] [ main] org.hibernate.orm.jpaa : HHH008540: Processing
PersistenceUnitInfo [name: default]
2026-08-02T08:26:09.623Z INFO 1 --- [ContanarizeSPMY] [ main] org.hibernate.orm.core : HHH000001: Hibernate ORM
core version 7.4.1.Final
2026-08-02T08:26:10.175Z INFO 1 --- [ContanarizeSPMY] [ main] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver
setup: ignoring JPA class transformer
2026-08-02T08:26:10.207Z INFO 1 --- [ContanarizeSPMY] [ main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-08-02T08:26:11.323Z WARN 1 --- [ContanarizeSPMY] [ main] org.hibernate.orm.jdbc.error : HHH000247: ErrorCode: 0,
SQLState: 08S01
2026-08-02T08:26:11.324Z WARN 1 --- [ContanarizeSPMY] [ main] org.hibernate.orm.jdbc.error : Communications link
failure
```

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the serv er.
2026-08-02T08:26:11.325Z WARN 1 --- [ContanarizeSPMY] [main] org.hibernate.orm.jdbc : HHH100046: Could not
obtain connection to query JDBC database metadata

org.hibernate.exception.JDBCConnectionException: Unable to obtain isolated JDBC connection [Communications link failure

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the serv er.] [n/a]
at org.hibernate.exception.internal.SQLStateConversionDelegate.convert(SQLStateConversionDelegate.java:80) ~[hibernate -
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.exception.internal.StandardSQLExceptionConverter.convert(StandardSQLExceptionConverter.java:34) ~[hibernate -
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.engine.jdbc.spi.SqlExceptionHelper.convert(SqlExceptionHelper.java:115) ~[hibernate -
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.engine.jdbc.spi.SqlExceptionHelper.convert(SqlExceptionHelper.java:101) ~[hibernate -
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.resource.transaction.backend.jdbc.internal.JdbcIsolationDelegate.delegateWork(JdbcIsolationDelegate.java:51)
~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
at
org.hibernate.engine.jdbc.env.internal.JdbcEnvironmentInitiator.getJdbcEnvironmentUsingJdbcMetadata(JdbcEnvironmentInitiator. java
:366) ~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.engine.jdbc.env.internal.JdbcEnvironmentInitiator.getJdbcEnvironment(JdbcEnvironmentInitiator.java:143)
~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.engine.jdbc.env.internal.JdbcEnvironmentInitiator.initiateService(JdbcEnvironmentInitiator.java:120) ~[hiber nate-
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.engine.jdbc.env.internal.JdbcEnvironmentInitiator.initiateService(JdbcEnvironmentInitiator.java:80) ~[hibern ate-
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.boot.registry.internal.StandardServiceRegistryImpl.initiateService(StandardServiceRegistryImpl.java:133)
~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.service.internal.AbstractServiceRegistryImpl.createService(AbstractServiceRegistryImpl.java:260) ~[hibernate -
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.service.internal.AbstractServiceRegistryImpl.initializeService(AbstractServiceRegistryImpl.java:235) ~[hiber nate-
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.service.internal.AbstractServiceRegistryImpl.getService(AbstractServiceRegistryImpl.java:212) ~[hibernate -
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.boot.model.relational.Database.<init>(Database.java:44) ~[hibernate -core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.boot.internal.InFlightMetadataCollectorImpl.getDatabase(InFlightMetadataCollectorImpl.java:251) ~[hibernate -
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.boot.internal.InFlightMetadataCollectorImpl.<init>(InFlightMetadataCollectorImpl.java:203) ~[hibernate -
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.boot.model.process.spi.MetadataBuildingProcess.complete(MetadataBuildingProcess.java:180) ~[hibernate -
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.jpa.boot.internal.EntityManagerFactoryBuilderImpl.metadata(EntityManagerFactoryBuilderImpl.java:1388)
~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
at
org.hibernate.jpa.boot.internal.EntityManagerFactoryBuilderImpl.populateSessionFactoryBuilder(EntityManagerFactoryBuilderImpl. java
:1468) ~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.jpa.boot.internal.EntityManagerFactoryBuilderImpl.build(EntityManagerFactoryBuilderImpl.java:1450)
~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
at
org.springframework.orm.jpa.vendor.SpringHibernateJpaPersistenceProvider.createContainerEntityManagerFactory(SpringHibernateJ pa
PersistenceProvider.java:93) ~[spring-orm-7.0.8.jar!/:7.0.8]
at
org.springframework.orm.jpa.LocalContainerEntityManagerFactoryBean.createNativeEntityManagerFactory(LocalContainerEntityManag
erFactoryBean.java:443) ~[spring-orm-7.0.8.jar!/:7.0.8]
at
org.springframework.orm.jpa.AbstractEntityManagerFactoryBean.buildNativeEntityManagerFactory(AbstractEntityManagerFactoryBean
.java:436) ~[spring-orm-7.0.8.jar!/:7.0.8]
at
org.springframework.orm.jpa.AbstractEntityManagerFactoryBean.afterPropertiesSet(AbstractEntityManagerFactoryBean.java:411)
~[spring-orm-7.0.8.jar!/:7.0.8]
at
org.springframework.orm.jpa.LocalContainerEntityManagerFactoryBean.afterPropertiesSet(LocalContainerEntityManagerFactoryBean. ja
va:419) ~[spring-orm-7.0.8.jar!/:7.0.8]
at
org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.invokeInitMethods(AbstractAutowireCapableBeanFa
ctory.java:1862) ~[spring-beans-7.0.8.jar!/:7.0.8]
at
org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.initializeBean(AbstractAutowireCapableBeanFactor y.j
ava:1811) ~[spring-beans-7.0.8.jar!/:7.0.8]
at
org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.doCreateBean(AbstractAutowireCapableBeanFactory
.java:603) ~[spring-beans-7.0.8.jar!/:7.0.8]
at
org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.createBean(AbstractAutowireCapableBeanFactory.ja
va:525) ~[spring-beans-7.0.8.jar!/:7.0.8]
at org.springframework.beans.factory.support.AbstractBeanFactory.lambda\$doGetBean\$0(AbstractBeanFactory.java:333) ~[spring -
beans-7.0.8.jar!/:7.0.8]
at org.springframework.beans.factory.support.DefaultSingletonBeanRegistry.getSingleton(DefaultSingletonBeanRegistry.java:371)
~[spring-beans-7.0.8.jar!/:7.0.8]
at org.springframework.beans.factory.support.AbstractBeanFactory.doGetBean(AbstractBeanFactory.java:331) ~[spring -
beans-7.0.8.jar!/:7.0.8]
at org.springframework.beans.factory.support.AbstractBeanFactory.getBean(AbstractBeanFactory.java:201) ~[spring -
beans-7.0.8.jar!/:7.0.8]
at
org.springframework.context.support.AbstractApplicationContext.finishBeanFactoryInitialization(AbstractApplicationContext.jav a:977)
~[spring-context-7.0.8.jar!/:7.0.8]

```
at org.springframework.context.support.AbstractApplicationContext.refresh(AbstractApplicationContext.java:621) ~[spring -
context-7.0.8.jar!/:7.0.8]
at
org.springframework.boot.web.server.servlet.context.ServletWebServerApplicationContext.refresh(ServletWebServerApplicationCon tex
t.java:143) ~[spring-boot-web-server-4.1.0.jar!/:4.1.0]
at org.springframework.boot.SpringApplication.refresh(SpringApplication.java:756) ~[spring-boot-4.1.0.jar!/:4.1.0]
at org.springframework.boot.SpringApplication.refreshContext(SpringApplication.java:445) ~[spring-boot-4.1.0.jar!/:4.1.0]
at org.springframework.boot.SpringApplication.run(SpringApplication.java:321) ~[spring-boot-4.1.0.jar!/:4.1.0]
at org.springframework.boot.SpringApplication.run(SpringApplication.java:1365) ~[spring-boot-4.1.0.jar!/:4.1.0]
at org.springframework.boot.SpringApplication.run(SpringApplication.java:1354) ~[spring-boot-4.1.0.jar!/:4.1.0]
at com.ContanarizeSpringMySQL.ContanarizeSPMY.ContanarizeSpmyApplication.main(ContanarizeSpmyApplication.java:10)
~[!/:0.0.1-SNAPSHOT]
at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method) ~[na:na]
at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(Unknown Source) ~[na:na]
at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(Unknown Source) ~[na:na]
at java.base/java.lang.reflect.Method.invoke(Unknown Source) ~[na:na]
at org.springframework.boot.loader.launch.Launcher.launch(Launcher.java:106) ~[app.jar:0.0.1-SNAPSHOT]
at org.springframework.boot.loader.launch.Launcher.launch(Launcher.java:64) ~[app.jar:0.0.1-SNAPSHOT]
at org.springframework.boot.loader.launch.JarLauncher.main(JarLauncher.java:40) ~[app.jar:0.0.1-SNAPSHOT]
Caused by: com.mysql.cj.jdbc.exceptions.CommunicationsException: Communications link failure
```

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the serv er.

```
at com.mysql.cj.jdbc.exceptions.SQLError.createCommunicationsException(SQLError.java:165) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.jdbc.exceptions.SQLExceptionsMapping.translateException(SQLExceptionsMapping.java:55) ~[mysql-connector-
j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.jdbc.ConnectionImpl.createNewIO(ConnectionImpl.java:840) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.jdbc.ConnectionImpl.<init>(ConnectionImpl.java:416) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.jdbc.ConnectionImpl.getInstance(ConnectionImpl.java:237) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.jdbc.NonRegisteringDriver.connect(NonRegisteringDriver.java:180) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.zaxxer.hikari.util.DriverDataSource.getConnection(DriverDataSource.java:144) ~[HikariCP-7.0.2.jar!/:na]
at com.zaxxer.hikari.pool.PoolBase.newConnection(PoolBase.java:373) ~[HikariCP-7.0.2.jar!/:na]
at com.zaxxer.hikari.pool.PoolBase.newPoolEntry(PoolBase.java:210) ~[HikariCP-7.0.2.jar!/:na]
at com.zaxxer.hikari.pool.HikariPool.createPoolEntry(HikariPool.java:488) ~[HikariCP-7.0.2.jar!/:na]
at com.zaxxer.hikari.pool.HikariPool.checkFailFast(HikariPool.java:576) ~[HikariCP-7.0.2.jar!/:na]
at com.zaxxer.hikari.pool.HikariPool.<init>(HikariPool.java:97) ~[HikariCP-7.0.2.jar!/:na]
at com.zaxxer.hikari.HikariDataSource.getConnection(HikariDataSource.java:111) ~[HikariCP-7.0.2.jar!/:na]
at
org.hibernate.engine.jdbc.connections.internal.DataSourceConnectionProvider.getConnection(DataSourceConnectionProvider.java:149)
~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.engine.jdbc.env.internal.JdbcEnvironmentInitiator
$ConnectionProviderJdbcConnectionAccess.obtainConnection(JdbcEnvironmentInitiator.java:508) ~[hibernate -
core-7.4.1.Final.jar!/:7.4.1.Final]
at org.hibernate.resource.transaction.backend.jdbc.internal.JdbcIsolationDelegate.delegateWork(JdbcIsolationDelegate.java:48)
~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
... 44 common frames omitted
Caused by: com.mysql.cj.exceptions.CJCommunicationsException: Communications link failure
```

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the serv er.

```
at java.base/jdk.internal.reflect.NativeConstructorAccessorImpl.newInstance0(Native Method) ~[na:na]
at java.base/jdk.internal.reflect.NativeConstructorAccessorImpl.newInstance(Unknown Source) ~[na:na]
at java.base/jdk.internal.reflect.DelegatingConstructorAccessorImpl.newInstance(Unknown Source) ~[na:na]
at java.base/java.lang.reflect.Constructor.newInstanceWithCaller(Unknown Source) ~[na:na]
at java.base/java.lang.reflect.Constructor.newInstance(Unknown Source) ~[na:na]
at com.mysql.cj.exceptions.ExceptionFactory.createException(ExceptionFactory.java:52) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.exceptions.ExceptionFactory.createException(ExceptionFactory.java:95) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.exceptions.ExceptionFactory.createException(ExceptionFactory.java:140) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.exceptions.ExceptionFactory.createCommunicationsException(ExceptionFactory.java:156) ~[mysql-connector-
j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.protocol.a.NativeSocketConnection.connect(NativeSocketConnection.java:79) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.NativeSession.connect(NativeSession.java:142) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.jdbc.ConnectionImpl.connectOneTryOnly(ConnectionImpl.java:964) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.jdbc.ConnectionImpl.createNewIO(ConnectionImpl.java:828) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
... 57 common frames omitted
Caused by: java.net.ConnectException: Connection refused
at java.base/sun.nio.ch.Net.pollConnect(Native Method) ~[na:na]
at java.base/sun.nio.ch.Net.pollConnectNow(Unknown Source) ~[na:na]
at java.base/sun.nio.ch.NioSocketImpl.timedFinishConnect(Unknown Source) ~[na:na]
at java.base/sun.nio.ch.NioSocketImpl.connect(Unknown Source) ~[na:na]
at java.base/java.net.SocksSocketImpl.connect(Unknown Source) ~[na:na]
at java.base/java.net.Socket.connect(Unknown Source) ~[na:na]
at com.mysql.cj.protocol.StandardSocketFactory.connect(StandardSocketFactory.java:144) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
at com.mysql.cj.protocol.a.NativeSocketConnection.connect(NativeSocketConnection.java:53) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
... 60 common frames omitted
```

```
2026-08-02T08:26:11.418Z WARN 1 --- [ContanarizeSPMY] [    main] org.hibernate.orm.deprecation      : HHH90000025:
MySQLDialect does not need to be specified explicitly using 'hibernate.dialect' (remove the property setting and it will be s elected by
default)
2026-08-02T08:26:11.432Z INFO 1 --- [ContanarizeSPMY] [    main] org.hibernate.orm.connections.pooling  : HHH10001005:
Database info:
Database JDBC URL [undefined/unknown]
Database driver: undefined/unknown
Database dialect: MySQLDialect
Database version: 8.0
Default catalog/schema: unknown/unknown
Autocommit mode: undefined/unknown
Isolation level: NONE [default NONE]
JDBC fetch size: undefined/unknown
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-08-02T08:26:12.202Z INFO 1 --- [ContanarizeSPMY] [    main] org.hibernate.orm.core              : HHH000489: No JTA platform
available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
2026-08-02T08:26:12.222Z INFO 1 --- [ContanarizeSPMY] [    main] com.zaxxer.hikari.HikariDataSource      : HikariPool -1 - Starting...
2026-08-02T08:26:13.226Z WARN 1 --- [ContanarizeSPMY] [    main] org.hibernate.orm.jdbc.error          : HHH000247: ErrorCode: 0,
SQLState: 08S01
2026-08-02T08:26:13.227Z WARN 1 --- [ContanarizeSPMY] [    main] org.hibernate.orm.jdbc.error          : Communications link
failure
```

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the serv er.

```
2026-08-02T08:26:13.235Z ERROR 1 --- [ContanarizeSPMY] [    main] j.LocalContainerEntityManagerFactoryBean : Failed to initialize
JPA EntityManagerFactory: Unable to build Hibernate SessionFactory [persistence unit: default]; nested exception is
org.hibernate.exception.JDBCConnectionException: Unable to open JDBC Connection for DDL execution [Communications link failur e
```

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the serv er.] [n/a]

```
2026-08-02T08:26:13.236Z WARN 1 --- [ContanarizeSPMY] [    main] ConfigServletWebServerApplicationContext : Exception
encountered during context initialization - cancelling refresh attempt: org.springframework.beans.factory.BeanCreationException: Error
creating bean with name 'entityManagerFactory' defined in class path resource
[org.springframework.boot/hibernate/autoconfigure/HibernateJpaConfiguration.class]: Unable to build Hibernate SessionFactory
[persistence unit: default]; nested exception is org.hibernate.exception.JDBCConnectionException: Unable to open JDBC Connection for
DDL execution [Communications link failure
```

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the serv er.] [n/a]

```
2026-08-02T08:26:13.243Z INFO 1 --- [ContanarizeSPMY] [    main] o.apache.catalina.core.StandardService  : Stopping service
[Tomcat]
2026-08-02T08:26:13.263Z INFO 1 --- [ContanarizeSPMY] [    main] .s.b.a.l.ConditionEvaluationReportLogger :
```

Error starting ApplicationContext. To display the condition evaluation report re-run your application with 'debug' enabled.

```
2026-08-02T08:26:13.279Z ERROR 1 --- [ContanarizeSPMY] [    main] o.s.boot.SpringApplication          : Application run failed
```

org.springframework.beans.factory.BeanCreationException: Error creating bean with name 'entityManagerFactory' defined in class path resource [org/springframework/boot/hibernate/autoconfigure/HibernateJpaConfiguration.class]: Unable to build Hibernate SessionFactory [persistence unit: default] ; nested exception is org.hibernate.exception.JDBCConnectionException: Unable to open JDBC Connection for DDL execution [Communications link failure

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the server.] [n/a]
at
org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.initializeBean(AbstractAutowireCapableBeanFactory.java:1815) ~[spring-beans-7.0.8.jar/:7.0.8]
at
org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.doCreateBean(AbstractAutowireCapableBeanFactory.java:603) ~[spring-beans-7.0.8.jar/:7.0.8]
at
org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.createBean(AbstractAutowireCapableBeanFactory.java:525) ~[spring-beans-7.0.8.jar/:7.0.8]
at org.springframework.beans.factory.support.AbstractBeanFactory.lambda\$doGetBean\$0(AbstractBeanFactory.java:333) ~[spring-beans-7.0.8.jar/:7.0.8]
at org.springframework.beans.factory.support.DefaultSingletonBeanRegistry.getSingleton(DefaultSingletonBeanRegistry.java:371) ~[spring-beans-7.0.8.jar/:7.0.8]
at org.springframework.beans.factory.support.AbstractBeanFactory.doGetBean(AbstractBeanFactory.java:331) ~[spring-beans-7.0.8.jar/:7.0.8]
at org.springframework.beans.factory.support.AbstractBeanFactory.getBean(AbstractBeanFactory.java:201) ~[spring-beans-7.0.8.jar/:7.0.8]
at
org.springframework.context.support.AbstractApplicationContext.finishBeanFactoryInitialization(AbstractApplicationContext.java:977) ~[spring-context-7.0.8.jar/:7.0.8]
at org.springframework.context.support.AbstractApplicationContext.refresh(AbstractApplicationContext.java:621) ~[spring-context-7.0.8.jar/:7.0.8]
at
org.springframework.boot.web.server.servlet.context.ServletWebServerApplicationContext.refresh(ServletWebServerApplicationContext.java:143) ~[spring-boot-web-server-4.1.0.jar/:4.1.0]
at org.springframework.boot.SpringApplication.refresh(SpringApplication.java:756) ~[spring-boot-4.1.0.jar/:4.1.0]
at org.springframework.boot.SpringApplication.refreshContext(SpringApplication.java:445) ~[spring-boot-4.1.0.jar/:4.1.0]
at org.springframework.boot.SpringApplication.run(SpringApplication.java:321) ~[spring-boot-4.1.0.jar/:4.1.0]
at org.springframework.boot.SpringApplication.run(SpringApplication.java:1365) ~[spring-boot-4.1.0.jar/:4.1.0]
at org.springframework.boot.SpringApplication.run(SpringApplication.java:1354) ~[spring-boot-4.1.0.jar/:4.1.0]
at com.ContainarizeSpringMySQL.ContanarizeSPMY.ContanarizeSpmyApplication.main(ContainarizeSpmyApplication.java:10)
~[/:0.0.1-SNAPSHOT]
at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method) ~[na:na]
at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(Unknown Source) ~[na:na]
at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(Unknown Source) ~[na:na]
at java.base/java.lang.reflect.Method.invoke(Unknown Source) ~[na:na]
at org.springframework.boot.loader.launch.Launcher.launch(Launcher.java:106) ~[app.jar:0.0.1-SNAPSHOT]
at org.springframework.boot.loader.launch.Launcher.launch(Launcher.java:64) ~[app.jar:0.0.1-SNAPSHOT]
at org.springframework.boot.loader.launch.JarLauncher.main(JarLauncher.java:40) ~[app.jar:0.0.1-SNAPSHOT]
Caused by: jakarta.persistence.PersistenceException: Unable to build Hibernate SessionFactory [persistence unit: default] ; nested exception is org.hibernate.exception.JDBCConnectionException: Unable to open JDBC Connection for DDL execution [Communications link failure

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the server.] [n/a]
at
org.springframework.orm.jpa.AbstractEntityManagerFactoryBean.buildNativeEntityManagerFactory(AbstractEntityManagerFactoryBean.java:448) ~[spring-orm-7.0.8.jar/:7.0.8]
at
org.springframework.orm.jpa.AbstractEntityManagerFactoryBean.afterPropertiesSet(AbstractEntityManagerFactoryBean.java:411) ~[spring-orm-7.0.8.jar/:7.0.8]
at
org.springframework.orm.jpa.LocalContainerEntityManagerFactoryBean.afterPropertiesSet(LocalContainerEntityManagerFactoryBean.java:419) ~[spring-orm-7.0.8.jar/:7.0.8]
at
org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.invokeInitMethods(AbstractAutowireCapableBeanFactory.java:1862) ~[spring-beans-7.0.8.jar/:7.0.8]
at
org.springframework.beans.factory.support.AbstractAutowireCapableBeanFactory.initializeBean(AbstractAutowireCapableBeanFactory.java:1811) ~[spring-beans-7.0.8.jar/:7.0.8]
... 22 common frames omitted
Caused by: org.hibernate.exception.JDBCConnectionException: Unable to open JDBC Connection for DDL execution [Communications link failure

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the server.] [n/a]
at org.hibernate.exception.internal.SQLStateConversionDelegate.convert(SQLStateConversionDelegate.java:80) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at org.hibernate.exception.internal.StandardSQLExceptionConverter.convert(StandardSQLExceptionConverter.java:34) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at org.hibernate.engine.jdbc.spi.SqlExceptionHelper.convert(SqlExceptionHelper.java:115) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at org.hibernate.engine.jdbc.spi.SqlExceptionHelper.convert(SqlExceptionHelper.java:101) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at
org.hibernate.resource.transaction.backend.jdbc.internal.DdlTransactionIsolatorNonJtaImpl.getIsolatedConnection(DdlTransactionIsolatorNonJtaImpl.java:71) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at
org.hibernate.resource.transaction.backend.jdbc.internal.DdlTransactionIsolatorNonJtaImpl.getIsolatedConnection(DdlTransactionIsolatorNonJtaImpl.java:37) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at
org.hibernate.tool.schema.internal.exec.ImprovedExtractionContextImpl.getJdbcConnection(ImprovedExtractionContextImpl.java:61) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at
org.hibernate.tool.schema.internal.exec.ImprovedExtractionContextImpl.getJdbcDatabaseMetaData(ImprovedExtractionContextImpl.java:68) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at
org.hibernate.tool.schema.extract.internal.InformationExtractorJdbcDatabaseMetaDataImpl.getJdbcDatabaseMetaData(InformationExtractorJdbcDatabaseMetaDataImpl.java:39) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at
org.hibernate.tool.schema.extract.internal.InformationExtractorJdbcDatabaseMetaDataImpl.processTableResultSet(InformationExtractorJdbcDatabaseMetaDataImpl.java:81) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at
org.hibernate.tool.schema.extract.internal.AbstractInformationExtractorImpl.getTables(AbstractInformationExtractorImpl.java:496) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at org.hibernate.tool.schema.extract.internal.DatabaseInformationImpl.getTablesInformation(DatabaseInformationImpl.java:124) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at
org.hibernate.tool.schema.internal.GroupedSchemaMigratorImpl.performTablesMigration(GroupedSchemaMigratorImpl.java:72) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at org.hibernate.tool.schema.internal.AbstractSchemaMigrator.performMigration(AbstractSchemaMigrator.java:220) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at org.hibernate.tool.schema.internal.AbstractSchemaMigrator.doMigration(AbstractSchemaMigrator.java:96) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at
org.hibernate.tool.schema.spi.SchemaManagementToolCoordinator.performDatabaseAction(SchemaManagementToolCoordinator.java:269) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at org.hibernate.tool.schema.spi.SchemaManagementToolCoordinator.lambda\$process\$1(SchemaManagementToolCoordinator.java:101) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at java.base/java.util.HashMap.forEach(HashMap.java:1000) ~[na:na]
at org.hibernate.tool.schema.spi.SchemaManagementToolCoordinator.process(SchemaManagementToolCoordinator.java:100) ~[hibernate-core-7.4.1.Final.jar/:7.4.1.Final]
at


```
org.hibernate.boot.internal.SessionFactoryObserverForSchemaExport.sessionFactoryCreated(SessionFactoryObserverForSchemaExport.java:35) ~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
    at org.hibernate.internal.SessionFactoryObserverChain.sessionFactoryCreated(SessionFactoryObserverChain.java:33) ~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
    at org.hibernate.internal.SessionFactoryImpl.<init>(SessionFactoryImpl.java:327) ~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
    at org.hibernate.internal.SessionFactoryRegistry.instantiateSessionFactory(SessionFactoryRegistry.java:64) ~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
    at org.hibernate.boot.internal.SessionFactoryBuilderImpl.build(SessionFactoryBuilderImpl.java:458) ~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
    at org.hibernate.jpa.boot.internal.EntityManagerFactoryBuilderImpl.build(EntityManagerFactoryBuilderImpl.java:1452) ~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
    at
    org.springframework.orm.jpa.vendor.SpringHibernateJpaPersistenceProvider.createContainerEntityManagerFactory(SpringHibernateJpaPersistenceProvider.java:93) ~[spring-orm-7.0.8.jar!/:7.0.8]
    at
    org.springframework.orm.jpa.LocalContainerEntityManagerFactoryBean.createNativeEntityManagerFactory(LocalContainerEntityManagerFactoryBean.java:443) ~[spring-orm-7.0.8.jar!/:7.0.8]
    at
    org.springframework.orm.jpa.AbstractEntityManagerFactoryBean.buildNativeEntityManagerFactory(AbstractEntityManagerFactoryBean.java:436) ~[spring-orm-7.0.8.jar!/:7.0.8]
    ... 26 common frames omitted
Caused by: com.mysql.cj.jdbc.exceptions.CommunicationsException: Communications link failure
```

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the server.

```
    at com.mysql.cj.jdbc.exceptions.SQLExceptionsMapping.translateException(SQLException.java:165) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.jdbc.exceptions.SQLExceptionsMapping.translateException(SQLException.java:55) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.jdbc.ConnectionImpl.createNewIO(ConnectionImpl.java:840) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.jdbc.ConnectionImpl.<init>(ConnectionImpl.java:416) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.jdbc.ConnectionImpl.getInstance(ConnectionImpl.java:237) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.jdbc.NonRegisteringDriver.connect(NonRegisteringDriver.java:180) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.zaxxer.hikari.util.DriverDataSource.getConnection(DriverDataSource.java:144) ~[HikariCP-7.0.2.jar!/:na]
    at com.zaxxer.hikari.pool.PoolBase.newConnection(PoolBase.java:373) ~[HikariCP-7.0.2.jar!/:na]
    at com.zaxxer.hikari.pool.PoolBase.newPoolEntry(PoolBase.java:210) ~[HikariCP-7.0.2.jar!/:na]
    at com.zaxxer.hikari.pool.HikariPool.createPoolEntry(HikariPool.java:488) ~[HikariCP-7.0.2.jar!/:na]
    at com.zaxxer.hikari.pool.HikariPool.checkFailFast(HikariPool.java:576) ~[HikariCP-7.0.2.jar!/:na]
    at com.zaxxer.hikari.pool.HikariPool.<init>(HikariPool.java:97) ~[HikariCP-7.0.2.jar!/:na]
    at com.zaxxer.hikari.HikariDataSource.getConnection(HikariDataSource.java:111) ~[HikariCP-7.0.2.jar!/:na]
    at
    org.hibernate.engine.jdbc.connections.internal.DataSourceConnectionProvider.getConnection(DataSourceConnectionProvider.java:149) ~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
    at org.hibernate.engine.jdbc.env.internal.JdbcEnvironmentInitiator$ConnectionProviderJdbcConnectionAccess.obtainConnection(JdbcEnvironmentInitiator.java:508) ~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
    at
    org.hibernate.resource.transaction.backend.jdbc.internal.DdlTransactionIsolatorNonJtaImpl.getIsolatedConnection(DdlTransactionIsolatorNonJtaImpl.java:44) ~[hibernate-core-7.4.1.Final.jar!/:7.4.1.Final]
    ... 49 common frames omitted
Caused by: com.mysql.cj.exceptions.CJCommunicationsException: Communications link failure
```

The last packet sent successfully to the server was 0 milliseconds ago. The driver has not received any packets from the server.

```
    at java.base/jdk.internal.reflect.NativeConstructorAccessorImpl.newInstance(Native Method) ~[na:na]
    at java.base/jdk.internal.reflect.NativeConstructorAccessorImpl.newInstance(Unknown Source) ~[na:na]
    at java.base/jdk.internal.reflect.DelegatingConstructorAccessorImpl.newInstance(Unknown Source) ~[na:na]
    at java.base/java.lang.reflect.Constructor.newInstanceWithCaller(Unknown Source) ~[na:na]
    at java.base/java.lang.reflect.Constructor.newInstance(Unknown Source) ~[na:na]
    at com.mysql.cj.exceptions.ExceptionFactory.createException(ExceptionFactory.java:52) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.exceptions.ExceptionFactory.createException(ExceptionFactory.java:95) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.exceptions.ExceptionFactory.createException(ExceptionFactory.java:140) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.exceptions.ExceptionFactory.createCommunicationsException(ExceptionFactory.java:156) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.protocol.a.NativeSocketConnection.connect(NativeSocketConnection.java:79) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.NativeSession.connect(NativeSession.java:142) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.jdbc.ConnectionImpl.connectOneTryOnly(ConnectionImpl.java:964) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.jdbc.ConnectionImpl.createNewIO(ConnectionImpl.java:828) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    ... 62 common frames omitted
Caused by: java.net.ConnectException: Connection refused
    at java.base/sun.nio.ch.Net.pollConnect(Native Method) ~[na:na]
    at java.base/sun.nio.ch.Net.pollConnectNow(Unknown Source) ~[na:na]
    at java.base/sun.nio.ch.NioSocketImpl.timedFinishConnect(Unknown Source) ~[na:na]
    at java.base/sun.nio.ch.NioSocketImpl.connect(Unknown Source) ~[na:na]
    at java.base/java.net.SocksSocketImpl.connect(Unknown Source) ~[na:na]
    at java.base/java.net.Socket.connect(Unknown Source) ~[na:na]
    at com.mysql.cj.protocol.StandardSocketFactory.connect(StandardSocketFactory.java:144) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    at com.mysql.cj.protocol.a.NativeSocketConnection.connect(NativeSocketConnection.java:53) ~[mysql-connector-j-9.7.0.jar!/:9.7.0]
    ... 65 common frames omitted
```

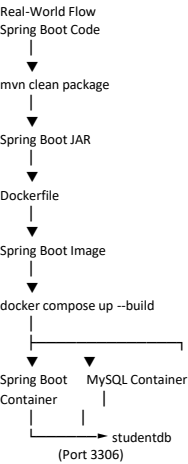
```
PS C:\Users\deepa\Documents\ContanarizeSPMY> docker logs mysql-container
2026-08-02 08:23:13+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.4.11-1.el9 started.
2026-08-02 08:23:14+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-08-02 08:23:14+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.4.11-1.el9 started.
2026-08-02 08:23:14+00:00 [Note] [Entrypoint]: Initializing database files
2026-08-02T08:23:14.412894Z [System] [MY-015017] [Server] MySQL Server Initialization - start.
2026-08-02T08:23:14.413808Z [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 8.4.11) initializing of server in progress as process 80
2026-08-02T08:23:14.424376Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-08-02T08:23:15.218168Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-08-02T08:23:18.558078Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please consider switching off the --initialize-insecure option.
2026-08-02T08:23:22.842989Z 0 [System] [MY-015018] [Server] MySQL Server Initialization - end.
2026-08-02 08:23:22+00:00 [Note] [Entrypoint]: Database files initialized
2026-08-02 08:23:22+00:00 [Note] [Entrypoint]: Starting temporary server
2026-08-02T08:23:22.902916Z 0 [System] [MY-015015] [Server] MySQL Server - start.
2026-08-02T08:23:23.207478Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.4.11) starting as process 125
2026-08-02T08:23:23.242539Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-08-02T08:23:24.474650Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-08-02T08:23:25.158293Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-08-02T08:23:25.158357Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
2026-08-02T08:23:25.166139Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
2026-08-02T08:23:25.216189Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.4.11' socket: '/var/run/mysqld/mysqld.sock' port: 0 MySQL Community Server - GPL.
2026-08-02T08:23:25.216277Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Socket: /var/run/mysqld/mysqld.sock
2026-08-02 08:23:25+00:00 [Note] [Entrypoint]: Temporary server started.
'var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
2026-08-02 08:23:28+00:00 [Note] [Entrypoint]: Creating database studentdata

2026-08-02 08:23:28+00:00 [Note] [Entrypoint]: Stopping temporary server
2026-08-02T08:23:28.177578Z 11 [System] [MY-013172] [Server] Received SHUTDOWN from user root. Shutting down mysqld (Version: 8.4.11).
2026-08-02T08:23:29.423117Z 0 [System] [MY-010910] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 8.4.11) MySQL Community Server - GPL.
2026-08-02T08:23:29.423149Z 0 [System] [MY-015016] [Server] MySQL Server - end.
2026-08-02 08:23:30+00:00 [Note] [Entrypoint]: Temporary server stopped
```

2026-08-02 08:23:30+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.

```
2026-08-02T08:23:30.197084Z 0 [System] [MY-015015] [Server] MySQL Server - start.
2026-08-02T08:23:30.512919Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.4.11) starting as process 1
2026-08-02T08:23:30.531913Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-08-02T08:23:31.751625Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-08-02T08:23:32.247527Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
2026-08-02T08:23:32.247676Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections
are now supported for this channel.
2026-08-02T08:23:32.257512Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the
path is accessible to all OS users. Consider choosing a different directory.
2026-08-02T08:23:32.303924Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.4.11' socket:
'/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
2026-08-02T08:23:32.303890Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: ':::port: 33060, socket:
/var/run/mysqld/mysqld.sock
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS          NAMES
fb22b9506ee8   mysql:8.4      "docker-entrypoint.s..." 41 minutes ago Up 41 minutes 0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp
mysql-container
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS          NAMES
a0dd9def7e84   containarizespmjy  "java -jar app.jar"      38 minutes ago Exited (1) 38 minutes ago
spring-container
fb22b9506ee8   mysql:8.4      "docker-entrypoint.s..." 41 minutes ago Up 41 minutes 0.0.0.0:3306->3306/tcp,
[::]:3306->3306/tcp mysql-container
e1439c60987c   nginx          "/docker-entrypoint...." 26 hours ago Exited (0) 26 hours ago
gifted_shtern
e5d68b44184a   nginx:stable-alpine3.24-perl "/docker-entrypoint...." 26 hours ago Exited (0) 26 hours ago
determined_buck
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS          NAMES
a0dd9def7e84   containarizespmjy  "java -jar app.jar"      About an hour ago Exited (1) About an hour ago
spring-container
fb22b9506ee8   mysql:8.4      "docker-entrypoint.s..." About an hour ago Up About an hour 0.0.0.0:3306->3306/tcp,
[::]:3306->3306/tcp mysql-container
e1439c60987c   nginx          "/docker-entrypoint...." 27 hours ago Exited (0) 27 hours ago
gifted_shtern
e5d68b44184a   nginx:stable-alpine3.24-perl "/docker-entrypoint...." 27 hours ago Exited (0) 27 hours ago
determined_buck
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker stop mysql-container
mysql-container
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker stop spring-container
spring-container
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker rm mysql-container
mysql-container
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker rm spring-container
spring-container
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docekr images
docekr : The term 'docekr' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ ~~~~~
+ docekr images
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (docekr:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException
```

```
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker images
i Info -> U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
containarizespmjy:latest 9ae31a1f5f69 347MB 0B
mysql:8.4 bced325a4ab7 813MB 0B
nginx:latest 4e5db4761e0f 161MB 0B U
nginx:stable-alpine3.24-perl 64788b97f1a5 103MB 0B U
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker rmi containarizespmjy
Untagged: containarizespmjy:latest
Deleted: sha256:9ae31a1f5f699203cf65209a7d976ddfd881453f524254f708c1126a84ae3c07
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker images
i Info -> U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
mysql:8.4 bced325a4ab7 813MB 0B
nginx:latest 4e5db4761e0f 161MB 0B U
nginx:stable-alpine3.24-perl 64788b97f1a5 103MB 0B U
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS          NAMES
e1439c60987c   nginx          "/docker-entrypoint...." 27 hours ago Exited (0) 27 hours ago gifted_shtern
e5d68b44184a   nginx:stable-alpine3.24-perl "/docker-entrypoint...." 27 hours ago Exited (0) 27 hours ago determined_buck
PS C:\Users\deepa\Documents\ContanarizeSPMY\ContanarizeSPMY>
```



Interview Tip

A very common interview question is:

Why do we use jdbc:mysql://mysql:3306/... instead of jdbc:mysql://localhost:3306/...

The answer is:

- localhost inside the Spring Boot container refers to the **Spring Boot container itself**, not your machine.
- Docker Compose creates an internal network and DNS.
- The service name (mysql) becomes the hostname, so Spring Boot reaches the MySQL container using mysql:3306.

This is the standard pattern used for Spring Boot applications deployed with Docker Compose.